ESE 3107 One-Shot Notes

Solar Thermal Engineering · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' notes (Khaja sir, Hasan sir) and the textbooks they follow (Duffie & Beckman, Kalogirou, G. N. Tiwari, Cengel)

One-shot notes: ESE 3107 Solar Thermal Engineering

Sun angles (A1)

$\delta = 23.45 \sin[360(284 + n)/365]$; $\omega = 15(ST - 12)$, morning −; $\cos\theta_z = \cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta$; $\alpha_s = 90 - \theta_z$; $\cos\gamma_s = (\cos\theta_z\sin\varphi - \sin\delta)/(\sin\theta_z\cos\varphi)$, sign of ω.

$\cos\omega_s = -\tan\varphi\tan\delta$; $N = 2\omega_s/15$ (derive from $\theta_z = 90°$). ST = clock + $4(L_{loc} - 90)$ + E (Bangladesh). Air mass $m = 1/\cos\theta_z$; $G_{sc} = 1367$ W/m^2.

Khulna Jan 18: δ −20.73, N 10.78 h. Khulna Mar 8, 11 AM: E −11.5 min ⇒ 10:47. Lat 45, Mar 5, 3 PM: sunrise 6:27, α 24.4°, γ_s 50.5°W, surface sunrise 7:22. Nirala: θ 21.4°, θ_z 39.0°, R_b 1.20 (use full cosθ formula).

Radiation and instruments (A2)

Extraterrestrial (outside atmosphere, beam only) vs terrestrial (beam + diffuse). $K_T = H/H_o$; diffuse fraction H_d/H falls as K_T rises (Liu-Jordan, Erbs).

$I_T = I_bR_b + I_d(1 + \cos\beta)/2 + I\rho_g(1 - \cos\beta)/2$; $R_b = \cos\theta/\cos\theta_z$; monthly $\bar{H}_T = \bar{R}\bar{H}$.

Pyranometer: thermopile under glass domes, global; shadow band ⇒ diffuse; beam = G − D.
Pyrheliometer: collimator tube on tracker, DNI.
Campbell-Stokes: glass sphere burns card, > 120 W/m^2.

Radiation laws (A3)

Blackbody: absorbs all, max emitter, diffuse. Planck $E_{b\lambda} = C_1/[\lambda^5(e^{C_2/\lambda T} - 1)]$; $E_b = \sigma T^4$; $\lambda_{max}T = 2898$ μm K. Gray: ε independent of λ. Kirchhoff: ε = α (enclosure equilibrium proof).

$E_b = \int\int I\cos\theta\sin\theta\, d\theta d\varphi = \pi I_b$. $d\omega = dA\cos\theta/r^2$.

2023: $E_{b\lambda}$(3 μm, 600 K) = 520, ω = 4.09×10^{-4} sr, Q = 2.74×10^{-4} W.

View factors and networks (A4)

Reciprocity $A_iF_{ij} = A_jF_{ji}$; summation ΣF = 1; superposition; crossed strings $F_{12} = (\Sigma\text{crossed} - \Sigma\text{uncrossed})/2L_1$. $F_{1\text{-}3} = F_{1\to(2,3)} - F_{1\text{-}2}$. 2019: F_{12} 0.075, F_{21} 0.050; 2022: 0.075.

Surface $R = (1 - \varepsilon)/A\varepsilon$, space $R = 1/AF$. Plates $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$ ⇒ 13.9 kW/m^2.

Reradiating $J = E_b$. Triangle duct: −28 kW/m (remove). N shields: $q/(N + 1)$.

Collectors (A5)

FPC (< 80 °C), ETC, CPC; trough, Fresnel (1-axis); dish, tower (2-axis). FPC: glazing, absorber (selective α 0.95, ε 0.1), risers + headers, insulation, casing. Greenhouse effect of glass.

ETC: vacuum kills convection/conduction ⇒ U_L 1-2 vs 4-8 ⇒ flatter efficiency line, better hot/cold/windy. Heat pipe: boil low, condense in manifold, drip back; tilt ≥ 25°; freeze-proof, one-way.

FPC performance (B1)

$Q_u = A_cF_R[S - U_L(T_i - T_a)]$; $\eta = F_R(\tau\alpha) - F_RU_L(T_i - T_a)/I$. $I_c = U_L(T_i - T_a)/(\tau\alpha)$. F′ (local fluid), F_R (inlet fluid) = F′F″.

$F' = (1/U_L)/\{W[1/U_L(D + (W - D)F) + 1/C_b + 1/\pi D_ih_{fi}]\}$; $F = \tanh x/x$, $x = m(W - D)/2$. $U_L = U_t + U_b + U_e$; Hollands Nu; $T_{sky} = T_a - 6$.

2020: F 0.961, F′ 0.912, F_R 0.868, Q_u 2.10 kW, η 65.7%. 2021: U_t 3.39 + U_b 0.6 = 4.0. 2019: U_t 5.40, U_L 6.23. 2022 dark test: 12.4.

Concentrators (B2)

$C = A_a/A_r$; acceptance $2\theta_c$; intercept γ. $C_{max} = 1/\sin^2\theta_s$ = 46,747 (3-D), $1/\sin\theta_s$ = 216 (2-D), θ_s = 0.265°: from $T_r \le T_s$.

Tracking: beam only, small acceptance angle. Single-axis (E-W, N-S, polar) vs two-axis. Pros: high T, small loss area. Cons: no diffuse, tracking, cleaning. Reflector: high specular ρ, durable, cheap.

Water heating and storage (B3)

Passive: thermosiphon (tank ≥ 0.3 m above collector), ICS (night loss: selective coat, covers). Active: direct (open loop, differential thermostat), indirect (antifreeze + heat exchanger). Reverse circulation: height, check valve.

$T^+ = T + [Q_u - L - UA(T - T_a)3600]/mc_p$. 2019: 45 → 86.4 °C. 2022: 40 → 116.9 °C (flag: above boiling, pressurised).

Applications (B4)

Dryers: direct (cabinet), indirect (collector + chamber + chimney), PV fans. Still: $q_{ew} = 0.016273h_{cw}(P_w - P_g)$, $P = e^{25.317 - 5144/T}$ Pa; 50/40 ⇒ 0.282 kg/m^{2}h; 60/50 ⇒ 335 W/m^2.

Desiccant wheel + heat wheel + evaporative cooler, regenerated at 60-90 °C. Absorption LiBr-water. Pond: 3 zones, salt gradient stops convection. Tower: heliostats, receiver, molten salt, turbine. Dish-Stirling 5-25 kW.

Definitions to write word for word

Solar constant: solar energy per unit time on a unit area normal to the rays at mean earth-sun distance outside the atmosphere, 1367 W/m^2. **Air mass**: path length of the beam through the atmosphere relative to the vertical path, $1/\cos\theta_z$. **Hour angle**: angular displacement of the sun east or west of the local meridian, 15°/h. **Declination**: angle of the sun at solar noon with the equatorial plane. **Zenith angle**: angle between the vertical and the sun line. **Solar azimuth**: angle from south to the horizontal projection of the sun line. **Sunshine hour**: time with beam irradiance > 120 W/m^2.

Clearness index: ground radiation / extraterrestrial radiation on a horizontal surface. **Diffuse fraction**: diffuse / global. **Geometric factor** R_b: beam on tilted / beam on horizontal. **Overall tilt factor**: total on tilted / total on horizontal.

Radiosity: all radiation leaving a surface (emitted + reflected). **Irradiation**: all radiation arriving. **Reradiating surface**: adiabatic, net radiation zero, $J = E_b$. **View factor**: fraction of radiation leaving i that strikes j directly. **Gray body**: ε independent of wavelength.

Critical radiation level: irradiance at which gain = losses. **F′**: actual gain / gain with absorber at local fluid temperature. **F_R**: actual gain / gain with whole absorber at inlet temperature. **Aperture**: opening receiving the sun. **Acceptance angle**: angular range accepted without moving. **Concentration ratio**: A_a/A_r. **Intercept factor**: fraction of focused beam hitting the receiver.

Sketches to practise (2 minutes each)

1. Sun angles on a horizon circle (θ_z, α_s, γ_s). 2. Pyranometer (domes, thermopile, shadow band) and pyrheliometer (tube on tracker). 3. Planck curves with Wien peaks. 4. Two-surface and three-surface (reradiating) networks. 5. FPC cross-section with labels (G-A-T-H-I-C). 6. Evacuated tube and heat-pipe tube. 7. Thermal network of FPC. 8. Efficiency line with $F_R(\tau\alpha)$ and slope. 9. Trough, Fresnel, dish, tower. 10. Thermosiphon loop with 0.3 m height. 11. Direct and indirect active systems. 12. Indirect dryer. 13. Solar still. 14. Salt-gradient pond. 15. Power tower with salt tanks. 16. Desiccant wheel chain.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
CPC	compound parabolic collector
CSP	concentrated solar power
DNI	direct normal irradiance
ETC	evacuated tube collector
FPC	flat plate collector
GHI	global horizontal irradiance
ICS	integrated collector storage
kWh	kilowatt-hour
LiBr	lithium bromide
LST	local standard time
MJ	megajoule
PV	photovoltaic
sr	steradian
ST	solar time
WMO	World Meteorological Organization